

REVIEW

Wearable micro-electro-mechanical systems pressure sensors in health care: Advancements and trends—A review

S. S. Kiran Kolluri¹ | S. Ananiah Durai² 

¹School of Electronics Engineering, VIT - Chennai, Chennai, Tamil Nadu, India

²Center for Nanoelectronics and VLSI Design, VIT - Chennai, Chennai, Tamil Nadu, India

Correspondence

S. Ananiah Durai, VIT Chennai, Kelambakkam-Vandalur Road, Rajan Nagar, Chennai, Tamil Nadu 00127, India.

Email: ananiahdurai.s@vit.ac.in

Abstract

Wearable technologies offer a complementary approach to clinical diagnostics by utilising a variety of physical, chemical, and biological sensors to mine physiological (biophysical and/or biochemical) data in real time (preferably continuous), in a non-intrusive or minimally invasive manner. Micro-Electro-Mechanical Systems (MEMS) pressure sensors dominate the healthcare applications especially for vital parameter sensing, as they feature the non-invasive method of diagnosis and have comparatively high sensitivity leading to better accuracy. Among them, capacitive and piezoresistive type pressure sensors have gained substantial advantages compared to other transduction devices due to high linearity, low power consumption, and low thermal coefficient. The performance review of such MEMS sensors in research and as well as market-ready devices that can be seamlessly integrated into commercial wearable products is the primary focus in this work. Challenges in the system level integration of Microsensors with the associated interface electronics and the design mitigation of such MEMS microsystems are also discussed. Design insights of analog front-end circuitry in terms of gain, noise, power and area that are crucial for any wearable applications are also comprehensively reviewed.

KEYWORDS

bioMEMS, microsensors, wearable devices

1 | INTRODUCTION

Wearable devices have become ubiquitous in the present-day technological world. These devices had become an integral part of our daily lives to the extent that their number has more than doubled in a span of just 6 years from 325 million in 2016 to 1.1 billion in 2022. It is predicted that the number of wearable devices may reach a whopping 2 billion in the near future. Most wearable technologies that facilitate real-time fitness and wellness tracking have become indispensable for the modern lifestyle. Further, such compact devices consisting of multiple microsensors are in recent days employed for daily self-monitoring of physical activities including gesture recognition, posture, rise and fall detection, strain at bone joints, step counting, gait pattern study, individual's angular position etc. Various remote medical diagnoses are also made possible through the state-of-the-art Body Area Network (BAN) in

which multiple wearable electronics are implemented to serve as sensor nodes to pick up the individual's vital parameters such as heart rate, blood oxygen saturation, blood pressure, electrocardiogram, ballistocardiogram etc. In spite of the advent of the modern technology, saving lives has become one of the most challenging tasks. According to the World Health Organisation, a major proportion of the deaths that occurred worldwide were because of the top 10 factors out of which few are ischaemic heart disease, heart stroke, chronic obstructive pulmonary disease, lower respiratory infections, trachea and lung cancers, Tuberculosis, diabetes etc. In most of the cases, continuous monitoring of patient's health data has become the need of the hour. This increasing demand fostered the burgeoning research and development of exceptional miniature sensors for fitness and diagnostic applications.

MEMS (Micro-Electro-Mechanical-System) sensors being the core part of the wearable devices debuted their presence in

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the early 1960s, wherein a single crystal silicon membrane pressure transducer is studied for electrical and mechanical properties [1] at Honeywell Research Centre, Hopkins. The results were at par with the predictions arrived by combining the isotropic nature of a uniformly loaded diaphragm and the piezoresistive properties of silicon material. Since then, researchers across the globe have significantly explored various other materials for developing wearable microsensors that can cater to medical diagnosis, wellness and fitness care. Though transduction principles such as capacitive, piezoresistive and piezo-electric [2, 3] were extensively explored, the former two became popular for bio-medical applications especially in the picking up of vital parameters that cater towards continuous health monitoring, owing to their performances. Recent advancements and the performance review of various such devices/systems are the primary focus of interest in this work that can provide a comprehensive overview of wearable microsystems for healthcare settings.

Section 2 of this paper describes few advanced wearable bio-medical MEMS sensors, which are in research and have the potential for commercialisation. Whereas, Section 3 describes various industry devices that are recently commercialised and are market available that meets the requirements of present-day health monitoring applications. Section 4 extensively provides key contributions of the signal processing circuitry. The Analog frontend that is analogous to the brain activity of the humans has been dealt right from the basic to the advanced multistage amplifiers. Section 5 deals with few critical challenges being faced in the design and deployment of various wearable technologies.

2 | WEARABLE MEMS SENSORS IN RESEARCH

With wearable MEMS gaining popularity day by day, meticulous and extensive research has been carried out by researchers around the world to make the technology further improve its capabilities in terms of various performance metrics through material choices, device dimensions and geometries. This section provides insights into the recently developed distinctive type of MEMS sensors implemented for the measurement of various human physiological parameters such as Blood Pressure (BP), respiration rate, heart rate etc. Blood pressure is a critical vital sign that impacts acute and long-term clinical decisions. Considering the significance in directing care, accurate and consistent blood pressure measurements utilising various MEMS devices that employ methods such as auscultatory [4, 5], oscillometry [6, 7], tonometry [8, 9], and cuff-less [10, 11] are explored.

A prototype of a wearable device developed for continuous Blood Pressure (BP) measurement using a MEMS 3-axis force sensor array was designed in ref. [12], as shown in Figure 1a. The triaxial adjustment mechanism of the sensor setup together with the onboard signal conditioning circuit enables calibration for the classical tonometry method of BP measurement even when the subject is under movement. The

uniqueness of this proposed setup lies in its capability of eliminating noise that is generated due to body movement, resulting in an artefacts-free pressure output. This has been achieved by employing a noise measurement module that includes a body movement noise generator and an error amplifier. The measured noise signal picked up through the sensor, which is indicative of the body movement, is deducted from the actual output of the BP measurement sensor by the error amplifier (pressure value + noise value). Therefore, the noise-free blood pressure of the person is extracted. The experimental analysis resulted in a precise blood pressure measurement with a standard deviation of 4.2 mmHg for the measurement range of 6.66–26.66 kPa (50–200 mmHg). Low noise capability of this BP measuring prototype as evident in Figure 1b is attractive for implementation in most wearables as the application demands increased precision while the user is in constant movement.

A single wearable device that can pick up multiple parameters rather than measuring and monitoring one specific parameter is cost-effective and user convenient, which otherwise requires utilising multiple sensors. A MEMS-based sensor with such intention was designed to measure at least two parameters such as the respiration rate and heart rate through pulse wave pick-up [13]. A single element of MEMS piezoresistive device, competent in measuring the aforesaid parameters, is utilised in a unique approach. This cantilever-based pressure sensor is the first of its kind that can be precisely placed in the nose pads of eyeglasses, in lieu of the conventional approach such as wrist and chest-based placement, to pick up the pulse wave from the angular artery as shown in Figure 2. When worn by the user, a thin silicone tube attached

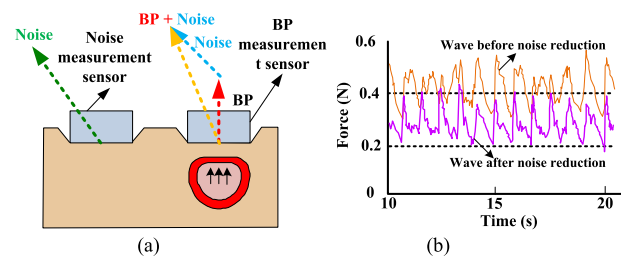


FIGURE 1 MEMS 3-axis force sensor (a) device structure (b) measurement results with and without noise [12].

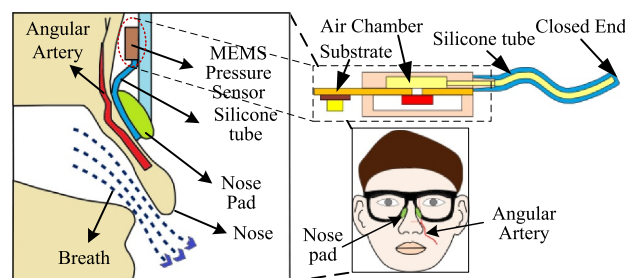


FIGURE 2 Cross-sectional view with an expanded image showing MEMS sensor device fitted with a silicone tube [13].

to the sensor chip meets the area between the nose pad and the nasal cavity and hence the underlying angular artery as shown in Figure 2.

Pressure change exerted due to inhalation/exhalation onto the device causes the displacement of the cantilever and hence variations in the resistivity of the piezoresistive device. This will result in a relative transduction that includes both low and high-frequency signals, which in turn enables the computation of the respiration and heart rate.

The experimental setup of the prototype was tested on a healthy male of age 34, and the resistance variations of the device recorded at a sampling rate of 1000 was as shown in Figure 3a. The analysis showed more fluctuations in resistance during breathing than during a pause, owing to the vibration caused by breathing, as shown in Figure 3b. The vibrations induced by respiration (both inhalation and exhalation) lie in the frequency range of 100–400 Hz. Hence, a high-pass filter with a cut-off frequency of 100 Hz has been employed to obtain the signal of interest. Also, an LPF with a cut-off frequency of 400 Hz has been included to extract the required pulse wave signal from the respiration of the person under test. As device placement has been the primary focus in this work with a significant amount of distance between the cantilever and the surrounding wall, the microsensor found to have the capability to measure differential pressures down to 0.1 Hz. Though the device yields better results in terms of Minimal Detectable Pressure (MDP), the physical stability of the eyeglasses was poor, thereby adding unwanted noise to the output signal, especially when the user is in movement. Also, the performance of the micro-sensor system degrades due to poor thermal coefficient as the transduction principle is of the resistive type. Capacitive pressure sensors featuring low noise, good thermal coefficient with high sensitivity that are built using biocompatible materials are in high demand, especially for wearable applications.

A MEMS pressure sensor with capacitance as the transduction principle is proposed in ref. [14] that is suitable for sensing various bio-parameter to overcome the drawback posed by [13]. The device includes a unique semi-major axis springs that strategically clamps the elliptic membrane only at the edges as shown in Figure 4, which rendered enhanced MDP than the design in ref. [13]. The poly-SiGe diaphragm hosed on top of the CMOS analog front-end (shown in Figure 5) forms a monolithically integrated CMOS-MEMS microsystem with the device dimension being $485 \times 280 \mu\text{m}$. A sensitivity of 1.7 mV/kPa

(@1.4 V supply) for pressure varying from 10 to 100 kPa has been reported with non-linearity being less than 1%. It is evident from the sensitivity and the dynamic range that this design can be utilised for various wearable biomedical applications that require better sensitivity. Further, flexible technology that might provide better MDP and sensitivity, recently gained much attention and popularity for wearable applications.

In ref. [15], a novel capacitive device with a distinctive flexible dielectric layer inspired by the origami structure is proposed as shown in Figure 6a. The flexible structure is a commendable attempt to increase the sensitivity when compared with the design in ref. [14]. It is evident that the composite dielectric layer developed with PDMS-doped

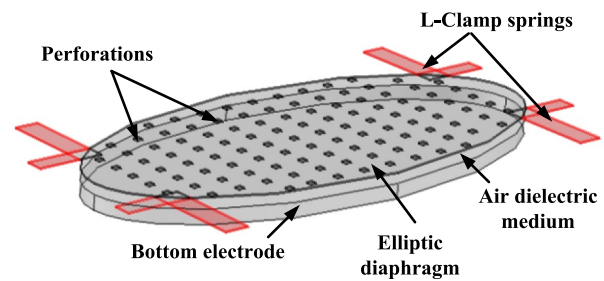


FIGURE 4 Elliptical capacitive MEMS device [14].

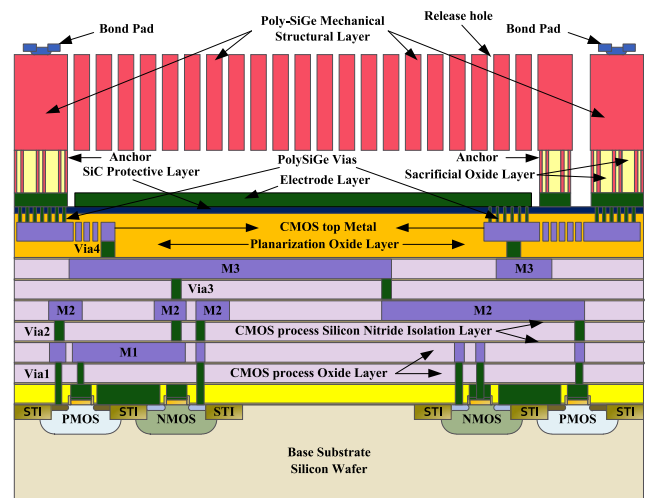


FIGURE 5 Cross-sectional view showing the MEMS device designed on top of the CMOS frontend [14].

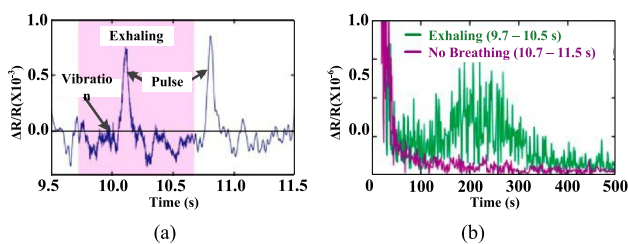


FIGURE 3 (a) Sensor's resistance variation, (b) frequency characteristics after FFT [13].

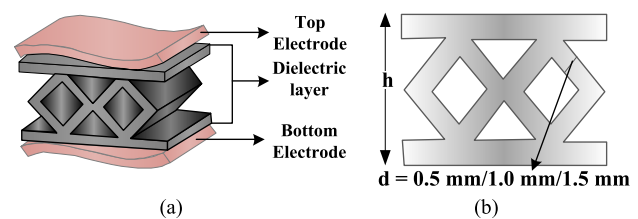


FIGURE 6 (a) Sensor with the origami structure (b) Cross-section of the sensor showing geometrical parameters [15].

BaTiO₃ and CNT, enhanced the sensitivity by achieving 0.1/kPa of relative capacitance variation with pressure. The sensor's performance has been demonstrated by placing it on the finger joint as a wearable to study the reflex movement of individuals. Alternatively, when placed at the knee joint where bending takes place rigorously when walking or climbing steps will enable in the diagnostic application of the walking pattern for gait analysis. The peak point of the output signal shown in Figure 7 corresponds to a bent joint, and the valley corresponds to a stretched joint. The sensor showed an excellent repeatability and sensitivity and has a high potential to be employed even in prosthetics.

Few other notable capacitive MEMS pressure sensors designed exclusively for Blood Pressure (BP) measurement have been reported recently [16–18] with a better dynamic range and sensitivity. Two such capacitive devices with square geometry have been demonstrated in ref.[16] for continuous monitoring of human BP. Silicon being the core material of the devices exhibited a wider dynamic range varying from almost vacuum to 50 kPa. A comparative FEM (Finite Element Analysis) analysis has been carried out for these two different sensors with 500 μm^2 (Design 1) and 1000 μm^2 (Design 2) being the surface area and having multiple thicknesses of 3 μm , 7 μm , 10 μm , 12 μm , and 15 μm . Various materials such as Silicon, Polysilicon and Silicon Carbide have also been considered for the comparative analysis to arrive at the right choice of the material that is feasible for implementation in wearables. Simulation results showed that the polysilicon diaphragm with a surface area of 500 μm^2 /1000 μm^2 for a thickness of 5 μm /15 μm and plate separation of 5 μm /3 μm yielded better capacitive sensitivity of 2.48 fF/kPa and 2.66 fF/kPa, respectively. Further, it is noted that another device having a similar device material but with circular geometry operating with higher linearity for the pressure range of 0–40 kPa has been proposed in ref. [17]. The device diaphragm having a diameter of 1000 μm exhibited a maximum capacitive sensitivity of around 12 fF/kPa for a diaphragm thickness of 15 μm and a plate separation of around 18 μm while under simulation. It can be inferred that the capacitive sensitivity of the circular diaphragms designed in ref. [17] is much better than the square diaphragms in ref. [16], and this is due to the fact that the area of the circular diaphragm is comparatively larger than the

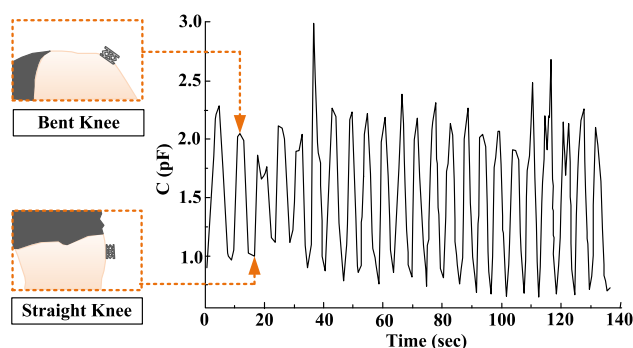


FIGURE 7 Performance of the device [15].

reported square diaphragms. Both the reported sensor devices in refs. [16, 17] can be suitably packaged to seamlessly integrate with off-chip interface electronics as a wearable microsystem.

In an attempt to achieve a comparatively larger dynamic range than demonstrated in refs. [16, 17] for better suitability in a wearable blood pressure measurement application, a relatively compact MEMS capacitive sensor was developed using Aluminium as the device diaphragm layer, with Si₃N₄ (Silicon Nitride) employed as the dielectric between the plates [18]. The functioning of this sensor follows the classical parallel plate capacitor mechanism, and the 3D structure depicting the layer stacks of the multi-material device is as shown in Figure 8. Nine different designs were demonstrated by the author with varied diaphragm thickness being 5 μm , 5.5 and 6 μm with the separation gap of 2 μm , 2.5 and 3 μm , respectively. In all these designs, the substrate material is considered as silicon with dimensions of 300 $\mu\text{m} \times 300 \mu\text{m} \times 70 \mu\text{m}$, whereas the Aluminium diaphragm and the bottom electrode's dimensions were optimised as 180 $\mu\text{m} \times 180 \mu\text{m}$ and 120 $\mu\text{m} \times 120 \mu\text{m}$, respectively. Sensitivity analysis performed in COMSOL Multiphysics revealed that among the nine designs, the diaphragm having a thickness of 5 μm and a separation of 2 μm exhibited the highest sensitivity of 0.0779 fF/kPa. Therefore, it is evident that the choice of such a miniaturised capacitive device in any wearable application having better sensitivity and linearity in the operating pressure range of 0–160 kPa is best suited for continuous blood pressure measurement. Though the capacitive devices in ref. [14, 16, 17] were found to have remarkable performances and are promising for wearable applications, packaging stress poses a serious threat, which might introduce a greater level of non-linearity affecting the sensitivity and the dynamic range of the devices. Therefore, desired packaging process steps that are specific to the wearable applications have to be strictly adhered to.

A wafer level self-packaged MEMS capacitive pressure sensor (CPS) designed and fabricated in ref. [19] as shown in Figure 9 will greatly decrease the possibility of performance degradation. This is achieved by optimising the device dimensions and 3D structure that will allow minimum consistent stress at the clamped edges. The designed touch mode operated MEMS capacitive device was vacuum sealed in a cavity using the Si-glass anodic bonding technique. With a sensitivity of 52.3 fF/kPa in the pressure range of 100–500 kPa, the CPS exhibited better linearity with an observed low non-linearity of

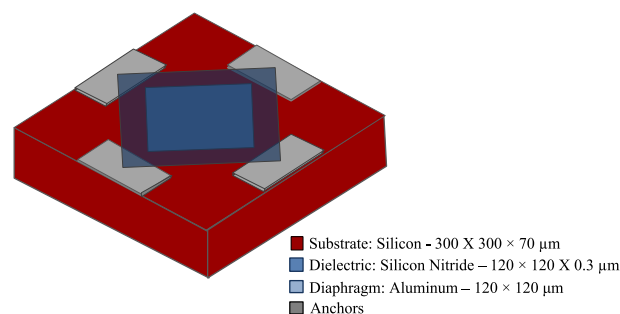


FIGURE 8 Cross-sectional view of the reported MEMS CPS [18].

2.7% FS. Though the demonstrated circular-shaped CPS has a larger radius of 2000 μm with a thickness being 40 μm , such a self-packaged design will be very much suitable for a miniaturised CPS microsystem for wearable applications. This high-performance device both in terms of sensitivity and linearity can suitably be implemented to monitor an individual's various physiological parameters.

The performances of other reported CPS designs [20–23] in terms of sensitivity, dynamic range, and compactness, with varied dimensions and geometries, were also found to be suitable for wearable applications. Sensing range versus diaphragm dimension comparison of the reviewed CPS devices [14–19] with other reported devices [20–23] that have notable sensitivity performance revealed that the elliptical device proposed by [14] showed larger sensitivity with a moderate dynamic sensing range. It can also be observed that the device in ref. [18] showed moderate sensitivity and the largest dynamic range, though the dimension was comparatively low, as shown in Figure 10. In Table 1, devices are categorised in terms of their geometries such as circular, square, and rectangular for fair comparison. Among the circular devices, the design in ref. [17] has a larger capacitance range that lies within the sensing range of vital parameters and remote BP monitoring applications. Thus, it proves to be a potential device to be developed as a product for wearable applications. The sensing range of both the devices of [16] appears to be suitable for ready implementation in wearables due to their maximum

capacitance variation under maximum applied pressure; however, their lower bound in capacitance will limit their utilisation.

3 | COMMERCIAL WEARABLE TECHNOLOGIES

Most researched and published work on MEMS pressure sensor devices has subsequently been developed as a prototype for clinical trials. Few such microsystems that yielded promising results have been successfully commercialised by a few leading microsystem manufacturers for utilization in healthcare applications. A wide range of commercial micro-sensor chips that are in the stage of prototyping and as well as are available as a product in the global market have been reviewed in this section. Further, there are few other products widely available as the micro-sensor system so as to provide end-to-end healthcare solutions for direct use by consumers. The attractive high-performance products have paved the way for effective implementation to read individual's vital parameters in most healthcare settings. Performances of such categories of devices and as well as products that have potential wearable applications are reviewed in the subsequent sub-sections.

3.1 | Microsensors for potential wearable applications

General purpose standalone MEMS pressure micro-sensor devices/systems are readily available in the market targeting both industrial and healthcare applications. Most popular devices are designed prominently employing either the piezoresistive or capacitive transduction technique. Devices that are suitable for various healthcare applications, such as vital parameter monitoring, fitness, sports, and medical diagnosis, have been comprehensively reviewed for their wearable compliance in the sub-sections below.

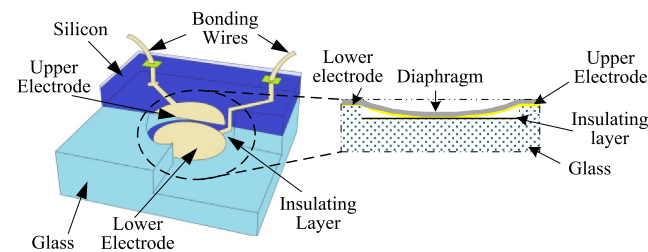


FIGURE 9 Wafer level self-packaged touch mode CPS [19].

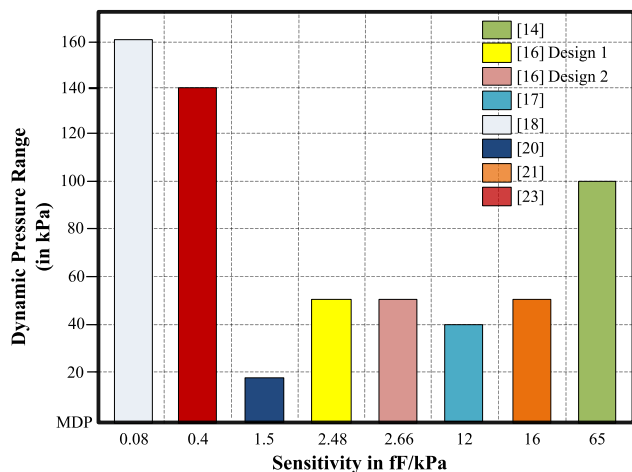


FIGURE 10 Sensitivity versus device dimensions with a varied dynamic range.

3.1.1 | Piezoresistive MEMS pressure sensor

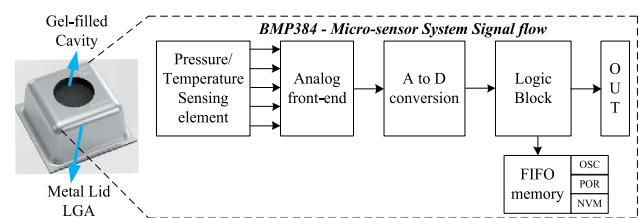
Bosch® Sensortec, a leader in bio-medical products, recently launched a compact microsensor (BMP384) that can measure pressure within the range suitable for various wearable applications. This BMP384 [24] features application-specific switching options between normal, forced, and sleep modes through a 2-bit mode register. Normal mode is preferred for applications requiring filtration of short-term disturbances through IIR (Infinite Impulse Response). However, in scenarios where only single measurement is required to be performed, force mode is invoked. This mode is mostly recommended for low-sampling rate applications. Once the measurement is recorded satisfactorily in any of the above modes, operations transit to sleep. As no measurements are usually performed in sleep mode, power consumption is negligible. The piezoresistive transduction-based sensor in

TABLE 1 Comparison of the reported device performance.

Device geometry	Reported work	Dimensions dia/area/thickness (μm)	Sensing range in kPa	Capacitance dynamic range in fF	Application
Circular	[20]	100/-/2	0–18	278–307.5	Bio-medical
	[23]	136/-/6	0–140	-	Bio-medical
	[17]	1000/-/15	0–40	0–550	Blood pressure
	[19]	2000/-/40	100–500	-	Bio-medical
Square	[22]	-/56 $\times$ 56/-	100–400	-	Blood pressure
	[18]	-/180 $\times$ 180/5.5	0–160	47.8–50.8	Blood pressure
	[16] design 1	-/500 $\times$ 500/3	0–50	110–235	Blood pressure
	[16] design 2	-/1000 $\times$ 1000/15		740–887	
Rectangular	[21]	-/130 $\times$ 258/4	0.2–50	0–160	PA pressure
Elliptical	[14]	-/485 $\times$ 280/4	10–100	120–440	Blood pressure

BMP384 has an on-board read-out interface that includes a mixed-signal ASIC CMOS readout that is capable of performing high precision A/D conversion for a seamless external interface. Housed in a metal-lid LGA package of size $2 \times 2 \times 0.95$ mm, as shown in Figure 11, the device with a gel-filled cavity exhibits improved robustness against water, chemicals, and other liquids. The operating range being 30–110 kPa with a moderate current consumption of $3.4 \mu\text{A}$ found to be attractive for battery-operated portable bio-medical applications. Further, this BMP384 as a system operating with an ultra-low supply voltage of 1.2 V is proven to be comparatively compact with a significantly high absolute accuracy of ± 50 Pa than other market-available devices such as BM1390GLV-Z [25]. Hence, the development of the remote health monitoring micro-sensor system that requires significant compactness, robustness and low power can employ BMP384 without any restrictions. One such application that demands such features is a remote spirometry monitoring wearable for individuals requiring remote monitoring of pulmonary functions.

Another compact product LPS22HB [26], being a digital nano-barometer, is found to be much more attractive for wearables than BMP384 [24] due to its significantly low thickness. Developed by yet another global high-tech semiconductor manufacturer, STMicroelectronics®, the core device in LPS22HB, has been designed with a suspended piezoresistive element for better sensitivity. A dedicated process that promisingly renders high yield resulted in the realisation of such a wider dynamic range absolute pressure sensor. Further, the high-resolution MEMS nanochip with a 24-bit data output consumes only a moderate current of $3 \mu\text{A}$, which is a critical requirement for any portable battery-operated wearable. The device exhibiting substantially low non-linearity in the pressure range of 26–126 kPa over the temperature range of -40°C – 85°C might attract global utilisation in applications requiring high linearity. A comparatively small and thin-holed LGA (HLGA) package with a footprint of $2 \times 2 \times 0.76$ mm makes the device attractive for various wearable applications, especially in activity/fitness monitoring.

**FIGURE 11** Bosch® BMP384 sensor [24].

In another exclusive wearable MEMS pressure sensor array designed and developed in ref. [27] for healthcare applications, Traditional Chinese medicine (TCM) pulse reading technique is employed. Unlike the classical pulse measuring technique of optical sensors embedded in wearable devices, an array of three commercial MEMS piezoresistive type digital barometers BMP085 [28] soldered on a PCB with a spacing of 13 mm is utilised to estimate the distance between each TCM pulse point. Artery-mimicking phantom arrangement with a flow pump, as shown in Figure 12, has been developed to carry out the physiologic that accurately reads a 1 Hz shear wave, as evident from the sine and carotid pulse wave plotted in Figure 13. The major downside of this design could be the pressure drop with increase in temperature due to resistive type transduction. The package rubber casting could be another cause for the poor thermal coefficient for which the sensor's onboard EEPROM may not be calibrated. In-house healthcare settings that maintain strict room temperatures might prefer such devices with poor thermal coefficients for wearable applications.

3.1.2 | Capacitive MEMS pressure sensor

A MEMS Digital Sensor for measuring barometric air pressure [29] that is capable of sensing temperature as well has been developed as a device-level product by the global semiconductor

leader, Infineon® Technologies. The capacitive sensing element employed in this product significantly improved the thermal stability than the piezoresistive device built micro-sensor systems in refs. [24, 26, 27], thus achieving a high precision of up to ± 0.0002 kPa, even during large temperature variations. The device exhibited a high accuracy of ± 0.1 kPa with a low current consumption of only 1.7 and 1.5 μA during pressure and temperature measurements, respectively. The device packaged with an 8-pin LGA has been found to have a minimal package size of $2 \times 2.5 \times 1$ mm, as shown in Figure 14. This compact DPS368, with an operating range of 30–120 kPa for pressure and -40°C – 85°C for temperature, includes an inbuilt-signal conditioning circuitry. Thus, the integrated micro-sensor system that does not

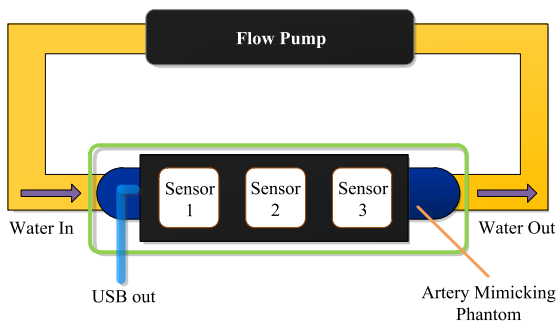


FIGURE 12 Experimental setup of the artery-mimicking phantom [27].

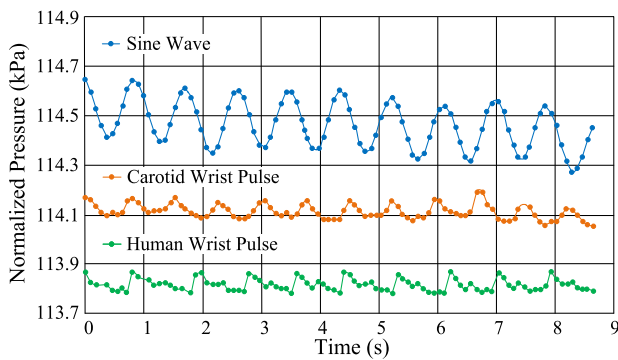


FIGURE 13 Preliminary human data with an artificial vertical offset to fit the three waveforms [27].

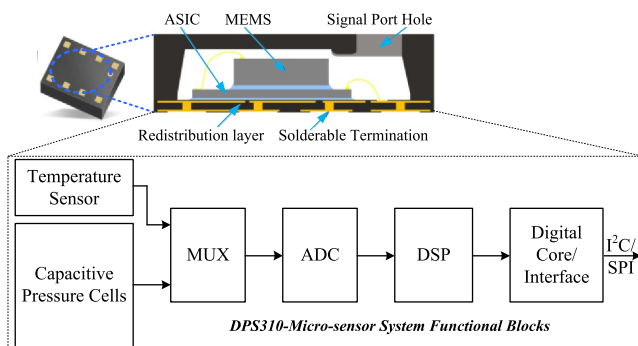


FIGURE 14 Infineon® DPS368 sensor [29].

require tedious off-chip interfacing of signal readout circuitry is found to be more attractive for any wearable application. The multiple read-out functional blocks together with an option of an output interface between I²C (Inter-Integrated Circuit) and SPI (Serial Peripheral Interface) can be seamlessly interfaced with an off-chip controller board for a complete microsystem realisation. DPS368 being comparatively compact with a low supply voltage of 1.7 V has significantly higher accuracy with ± 0.1 kPa over the wider temperature range than other commercial pressure sensors such as MS56702BA03 [30]. Thus, the wider sensing range together with its high accuracy and better thermal stability makes it suitable for employment in health monitoring/fitness tracking applications.

An exclusive MEMS prototype to pick up the real-time human vital parameters from the pulse wave has been reported in ref. [31]. It employs an array of MEMS pressure sensors to achieve relatively high resolution. The developed prototype that utilises Murata's SCB10H MEMS capacitive pressure sensor is reported to pick the arterial waveform for absolute pressure measurement over the maximum capacitance variations of up to 4 pF, with a maximum pressure being 120 kPa. The device exhibits minimal material stress and maximised overload performance with a significantly high sensitivity of 55 fF/kPa @ 100 kPa. The compact sensor array of dimensions $1.4 \times 1.4 \times 0.85$ mm is housed inside a gel-type material for better coupling. Assembled on a flexible wristband that can be comfortably wrapped around the wrist, it should be ensured that there is sufficient physical contact between the device and the region of the radial artery. The sensor capacitance variation due to the pressure exerted by the radial artery pulse has been picked up by the PCap04Z (Make: PMT, Florida), a capacitance-to-digital converter. This weak signal is then processed by EFM32, a 32-bit ARM Cortex M3 micro-controller (Make: Silicon Labs), for better signal integrity. Then, this signal is acquired by the Personal Computer (PC) through a serial port by means of a python application for subsequent processing and classification. After a series of steps such as decimation, band-pass filtering etc., the AMPD (Automatic Multiscale-based peak Detection) algorithm is applied [32] to obtain the peaks for the separation of noise from the actual peaks through adaptive thresholding. Further steps of waveform averaging and analysis resulted in the extraction of dichotic notch, systolic, and diastolic peaks computed from the signal derivatives. The relationship of these parameters with cardiovascular health that were reviewed in ref. [33] proves the suitability of the implementation of prototypes in health monitoring devices. Hand, body movement, and muscle tension might hamper the amplitude variations of the transduced signal, which in turn might lower the sensitivity of the device. Therefore, addressing such concerns through proper placement is essential.

Among the reviewed general purpose MEMS microsensor, four of the best products in terms of compactness and accuracy that can have potential applications in health care as wearable have been compared in Table 2. Murata's SCB10H MEMS capacitive pressure sensor appears to be more promising due to its superior compactness and moderate accuracy.

TABLE 2 Potential Commercial Products comparison.

Parameters	Infineon® DPS368 [29]	Murata® SCB10H [31]	Bosch® BMP384 [24]	ST® LPS22HB [26]
Sensor type	Digital barometric pressure sensor	Absolute pressure sensor element	Digital pressure sensor	Absolute digital output barometer
Interface	I ² C and SPI	External	I ² C and SPI	I ² C and SPI
Sensing principle	Capacitive	Capacitive	Piezo-resistive	Piezo-resistive
Accuracy	0.1 kPa	0.04 kPa	0.05 kPa	0.01 kPa
Pressure range	30–120 kPa	30–120 kPa	30–110 kPa	26–126 kPa
Dimensions (in mm)	2.0 × 2.5 × 1.1	1.4 × 1.4 × 0.85	2 × 2 × 0.95	2 × 2 × 0.76
Potential applications	Fall detection, air flow monitoring, smart inhaler	Medical devices	Health care (spirometry)	Fitness tracking

Whereas if accuracy is a major constrain in an application, LPS22HB would be a better choice. BMP384, on the other hand, provides comparatively moderate accuracy with moderate compactness.

3.2 | Wearable MEMS products in market

Most market-available wearable sensors have not yet been found in widespread usage by both healthcare centres, as well as by common individuals, due to technological and social challenges. Technological challenges such as operating sensing range and its associated transduction parametric range, accuracy deviations with and without body movement, bulky devices causing uncomfortable placement for long-term wearing and delayed/unreliable measurement must be addressed. Recent attempts to overcome such challenges by few market players are reviewed in the subsequent sections.

3.2.1 | Multi-element MEMS pressure sensor

The mounting yield of any conventional multi-element MEMS pressure sensors deteriorates due to multiple electrodes that extract signals, and as well as due to bulky electronics that are designed to enhance noise resistance. Further, the necessity of active resin for continuous health monitoring applications, especially in Blood Pressure measurement, makes it unsuitable for wearable applications. To address the above challenges in such beat-by-beat blood pressure measurement, OMRON® HEALTHCARE Co. Ltd. has designed a novel MEMS pressure sensor with 46 sensing elements integrated with Application Specific Integrated Circuits (ASIC) [34]. The limitations such as non-mobility and resting position while signal readout that prevails in the classical methods such as Oscillometry and tonometry have also been overcome. The multi-element MEMS pressure sensor device shown in Figure 15 has a length of 11.5 mm with a width being 3.0 mm and a thickness of 400 µm. A comparatively compact device that is developed to suit various wearable BP measurements has diaphragm dimensions of 9.5 mm × 0.26 mm × 5 µm. New packaging and assembly methods have been adapted to preserve the highest

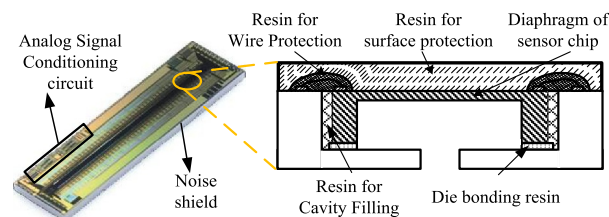


FIGURE 15 Multi-element MEMS pressure sensor chip [34] (OMRON® HEALTHCARE Co. Ltd.).

precision, production yield and to provide long-term wearing comfort. The utilisation of such application-specific MEMS device in wearables will enhance the widespread usage not only by the critically ill individuals but also by the common public.

Portable Sphygmomanometer equipped with the typical Multi-element MEMS pressure sensor comprises of 46 piezoresistive bridge circuits placed at short intervals of 200 µm that were formed on top of the diaphragm to achieve compactness. The hybrid micro-sensor system is integrated with the advanced ASIC circuitry that includes multiplexer, operational auscultate and filter stages for lossless signal readout. The typical setup of the sphygmomanometer fitted on to the wrist of human hand is shown in Figure 16a that can directly display the blood pressure digitally on to the OLED display. Figure 16b shows the simulation plot of sensor's pressure output matched with that of the pressure waveform of the blood vessel. The devices exhibited a high determination coefficient in the optimum pressing condition. The superior static pressure–temperature properties and the accuracy of the pressure sensor were reported to be comparatively high.

3.2.2 | MEMS pressure sensor for sleep apnea detection

AIDMED® One recorder [35], a compact commercial medical device with multi-functional features, has been designed to pick up and analyse the patients' physiological parameters through various modules. The differential pressure sensor device being one of the core modules is utilised to perform

remote continuous measurement of an individual's respiratory rate. The airflow through the nose can be measured using the intranasal cannula to observe the sleep-disordered breathing and to assess the relationship of the recorded heart rhythm with the depth of breaths. The MEMS sensor employs a piezoresistive effect to measure the variable pressure exerted by the airflow on to a silicon or polysilicon diaphragm, which then are processed by an interface of electronics to obtain an output voltage proportional to the applied pressure. The sensed parameters serve as the marker for the detection of sleep apnoea to determine the level of chronic obstructive pulmonary disease (COPD) followed by tele-rehabilitation. Measurement waveforms to detect the condition of sleep apnoea in a clinical trial are shown in Figure 17. The geometrical shape and the dimensions of the pressure sensor were the primary focus to optimise and address the technological challenges to achieve global widespread device usage by individuals. The system level precision, accuracy, and durability of the wearable are reported to be satisfactory with the high quality of service.

4 | REVIEW OF ANALOG FRONTEND CIRCUITS

A signal conditioning module popularly referred to as interface electronics in any micro-sensor system requires stringent design techniques as it determines the overall performances. It plays a vital role in picking up the feeble, weak and noisy analog signal obtained through the various transduction technique employed by the microsensor devices, without any attenuation. It converts the informative signal to be compatible with the

subsequent process/monitoring or control modules. Such read-out circuits are considered to be intelligent and crucial constituents of any wearable micro-sensing system that is analogous to the human brain processing of bio-signals from any sensory organ. This section discusses in detail the various signal conditioning circuits that cater to the improvement of the overall performances of the wearable such as system level sensitivity, low power, high gain, and better linearity.

The waveform profile of the unprocessed signal received from the wearable microsensors is unreliable and not beneficial enough to directly drive the subsequent stages of the sensor system, as it is weak and usually superimposed with noise. Various techniques such as chopper modulation and demodulation [36], auto-zeroing [37], correlated double sampling, and switched capacitor circuits [38] are designed to eliminate the non-ideal behaviour of the circuit components that will minimise/eliminate the noise. In addition to noise immunity, several other critical parameters including linearity, stability, and resolution are of significant concern. However, power dissipation is predominantly regarded as the parameter of paramount importance in any read-out electronics design, particularly for wearable microsystems where the extended battery life is essential for continuous health monitoring without interruption. This review encompasses various reported read-out electronics specifically designed for sensor signal conditioning, evaluating their performance and suitability for implementation in wearable microsystems.

A signal conditioning circuit specifically designed and analysed in ref. [39] for MEMS accelerometer signal readout has a simplistic design style with low circuit complexity. This interface electronics, comprising only a few functional blocks as illustrated in Figure 18, demand relatively minimal power. The utilisation of elementary circuit stages such as modulator, buffer, amplifier, demodulator, and low pass filter enabled the overall system level power reduction. Two distinct demodulation schemes such as single- and double-switch demodulators were proposed as depicted in Figures 19 and 20 respectively, to cater for application-specific implementation. The circuits were designed and simulated using UMC 180 nm CMOS technology within the Cadence Virtuoso tool environment. Results indicated that the double-switch demodulator outperformed the single-switch counterpart in terms of noise and offset performances. Specifically, noise reduction was nearly doubled with the double-switch scheme compared to the single-switch

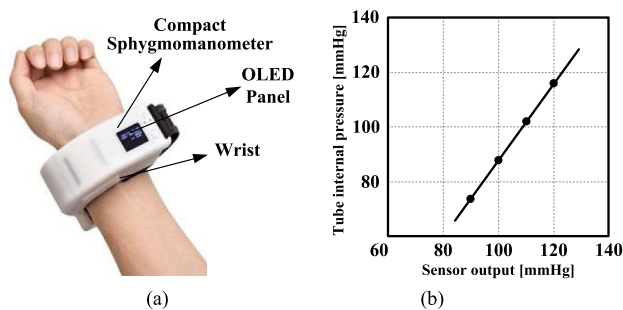


FIGURE 16 (a) Sphygmomanometer equipped with multi-element sensor, (b) internal pressure versus sensor output [34].

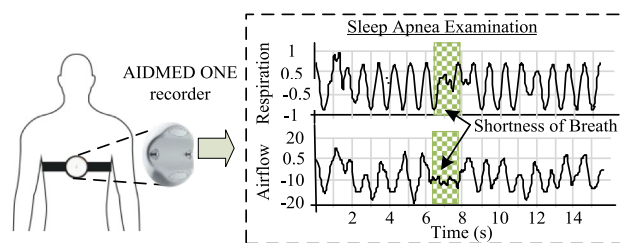


FIGURE 17 AIDMED[®] one recorder placed on the human body [35].

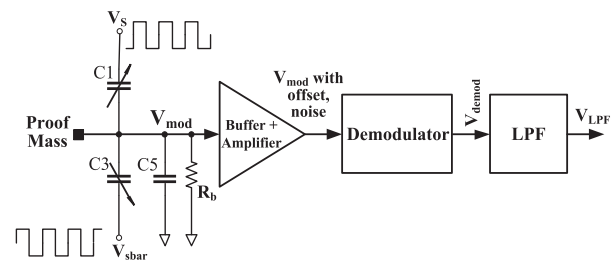


FIGURE 18 Interface electronics for the accelerometer application [39].

variant; however, only a marginal increase in dynamic power is observed due to an additional switching element. Further, it is evident from Figure 21 that the offset offered by single-switch demodulator is significantly high compared to the double-switch demodulator. Therefore, it is preferable to opt for the later design in any wearable applications, where noise and offset performances are crucial. Though the topology designed in the later might initially appear bulky due to an additional switch, comparatively fewer components have invariably lowered the chip area; hence, it proves to be an attractive choice to design compact wearables.

In an attempt to enhance the signal strength to maintain integrity, while reducing the overall power budget, an on-chip signal conditioning circuit is proposed for the capacitive pressure sensor in ref. [40]. In such micro-sensor devices, the capacitance variation due to the exerted pressure on the diaphragm, causes a proportionate variation in current. This single-ended current processed by an initial trans-impedance pre-amplifier causes a single to differential current to voltage conversion. The differential conversion is essential at this stage for effective noise cancellation at the subsequent fully

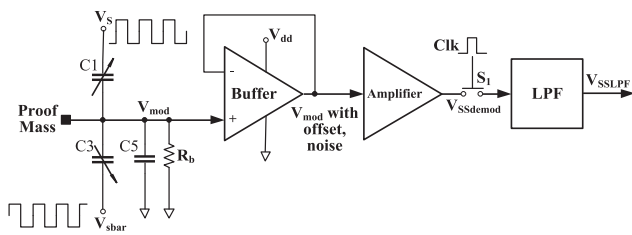


FIGURE 19 Single-switch demodulator-based interfacing circuit [39].

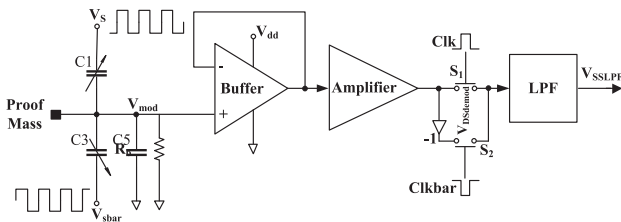


FIGURE 20 Double-switch demodulator-based design [39].

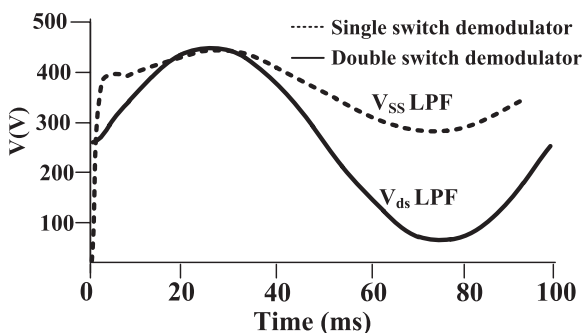


FIGURE 21 Filtered signal of the demodulator [39].

differential high gain stage. This high gain operational amplifier (Opamp) incorporates a transmission gate based chopper stabiliser to further minimise the $1/f$ noise eventually, yielding a lower noise floor of $100 \text{ nV}/\sqrt{\text{Hz}}$. The frequency compensation and stability of the CMOS circuitry were also improved through pole splitting and pole pushing techniques. A 4th-order low pass filter has then been cascaded at the output of the chopper stabilised Opamp to achieve sharp roll off that effectively removes the chopper residuals. An active resistance matching to achieve $1/gm_5$ is employed for minimising the gain error that would otherwise degrade the signal. The final signal-conditioned sensor output should be capable of driving an off-chip Analog to Digital Circuit (ADC) for display or actuation purposes; therefore, high output impedance prevailing at the output of the filter is not preferable.

A self-biased differential buffer that features low output impedance and better linearity was opted in this work. This output buffer stage has not only provided the required driving capability but also low area and power. The elimination of multiple bias circuitry enabled to achieve low power dissipation and area, which are essential for wearable application. Although conventional source followers can be the best choice due to its reduced design complexity, non-zero offset and high non-linearity render it as unsuitable for critical applications such as the wearable health monitoring system.

An ultra-low power signal conditioning circuit shown in Figure 22 can be a better choice as read-out electronics for direct implementation in most wearable MEMS microsystems [41]. The two-stage design with an initial block being a current amplifier that substantially reduces noise floor with low power dissipation appears to be an attractive input stage for a precise signal pick-off in any continuous health monitoring devices. The subsequent stage comprises two parallel paths, enhancing the Signal-to-Noise Ratio (SNR) with adjustable dual gain mode settings. Furthermore, the amplifiers in parallel pathways were modulated at two distinct frequencies to alleviate flicker noise.

The integrated input-referred noise voltage of the dual chopper amplifier operated in both Single-Path Mode (SPM) and Dual-Path Mode (DPM) with multiples of 500 Hz frequency ranges is shown in Figure 23. It is evident from Table 3 that the dual-path mode exhibited significantly high SNR compared to the single-path mode, for a similar signal amplitude in the operating frequency range of 0.5–4 kHz. The demonstrated modified chopper amplifier that features significantly low noise,

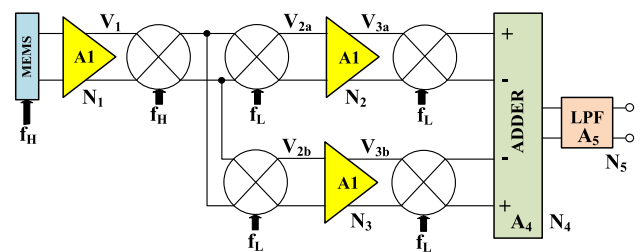


FIGURE 22 Modified dual chopper amplifier [41].

high SNR and power dissipation as low as $2.66 \mu\text{W}$ for a 0.7 V supply appears to be comparatively a better choice for signal readout in wearables.

The critical challenge in achieving a high performance multistage signal conditioning module lies in the choice and the design of the high-gain amplifier with good stability, linearity, and reduced power consumption. Recently, the Recycled Folded Cascode (RFC) design is popularly adopted as the core amplifier in the microsensor read out [42]. The primary advantage lies in the fact that the recycling current technique yielded better gain enhancement even with a single differential stage. This eliminates the need for utilising multiple amplifier stages to achieve the required gain as opposed to the design in ref. [41]. In such designs, the utilisation of a few number of transistors not only leads to a substantial reduction in the chip area but also a significant reduction in overall power consumption is achieved.

The conventional folded cascode amplifier exhibited poor transconductance due to the low signal current through the folding node transistors. This has been overcome with the recycling current concept that significantly increases current through the node, thereby invariably increasing the transconductance and hence the gain. This has been achieved by splitting the input devices without utilising additional transistors. The amplifier implemented in UMC 180 nm CMOS technology exhibited a high single-stage gain of 62.3 dB with a unity gain bandwidth of 53.31 MHz. In addition, a novel technology independent biasing employed in ref. [42], referred as the potential distribution method (PDM), enabled in yielding high linearity with a phase margin of 73.4° . Thus, it can be concluded that the designed RFC circuit shown in Figure 24 yielded substantially low noise, power, area and better linearity. The area reduction evident from the device dimensions in Table 4 proves its compactness and suitability of employment in wearable applications.

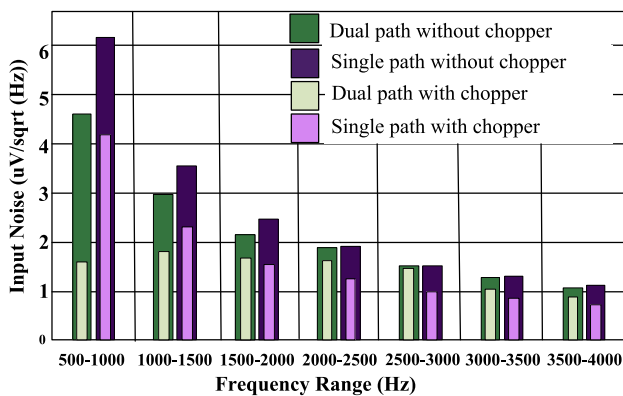


FIGURE 23 Input-referred noise of the dual chopper amplifier [41].

TABLE 3 Dual chopper amplifier operation mode [41].

Input voltage	Mode	Gain (dB)	SNR (dB)	Frequency range (kHz)
1.16 mV	SPM	34	89.06	0.5–4
	DPM	38	92.85	

The refinement of the basic RFC circuit through the integration of additional transistors M_{3c} and M_{4c} , as shown in Figure 25 [43], has resulted in further enhancement in gain.

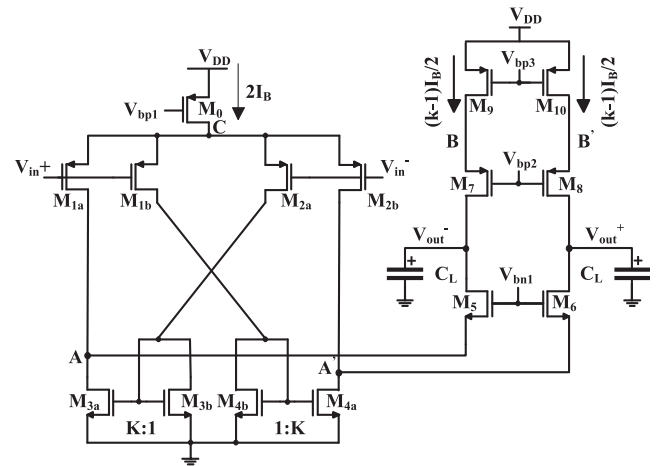


FIGURE 24 Recycling folded cascode amplifier [42].

TABLE 4 Device aspect ratios for $L = 0.5 \mu\text{m}$ [42].

Devices	Width in μm
M_0	87.26
$M_{1a}, M_{1b}, M_{2a}, M_{2b}$	3.72
M_{3a}	39.9
M_{3b}	13.32
M_{4a}	39.9
M_{4b}	13.32
M_5, M_6	2.35
M_7, M_8	36.2
M_9, M_{10}	26.13

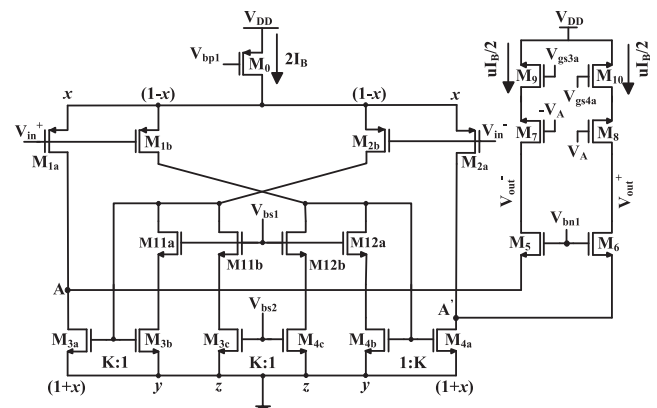


FIGURE 25 I-RFC amplifier [43].

This Improved RFC (I-RFC) features separate pathways for AC and DC currents, thereby augmenting transconductance. However, the inclusion of additional current recycling devices such as M_{11a} , M_{11b} , M_{12a} , and M_{12b} , and also the signal incrementing devices such as M_{2b} and M_{1b} , has adversely affected the area performances. Despite its inferior area performance compared to the basic RFC, there is only a minimal deviation in power dissipation between I-RFC and [42], due to the multi-recycling loop that reuses both AC and DC currents efficiently. Consequently, since both designs described in refs. [42, 43] demand comparable power budgets, either can be selected for wearable applications unless otherwise area is a critical constraint.

A Quadruply-split (QS) cross-driven doubled RFC amplifier is proposed in ref. [44] to further enhance the gain within the desired power budget. The use of four fractionally split input differential pair transistors with appropriate aspect ratios significantly increased the transconductance and single-stage gain, without compromising the UGB (Unity Gain Bandwidth) value. Furthermore, to increase small signal current and hence the gain, specific differential mirroring ratios (F and H) are employed using dual splitting current mirrors, as illustrated in Figure 26. This has emanated in almost doubled

transconductance without significantly increasing the area and power. In due course, the splitting of bias current resulted in faster settling time and better slew rate. The amplifier designed in the 130 nm CMOS technology of the IBM process demonstrated a significantly low noise level of $48.3 \mu V_{rms}$ with an increased single-stage gain of 76 dB and better phase margin. The proposed fully differential circuit with a differential difference common mode feedback that can be seamlessly integrated with any microsensor has comparatively better performances than earlier work. It is evident from the above post-fabricated experimental results of this single-chip CMOS + MEMS pressure sensor that such advanced design, though it appears to be complex, will promisingly meet the low power and noise requirements for utilisation in a distinct wearable micro-sensor system. Also, the comparison results in Table 5 proves that the QS-RFC has provided better gain, slew rate and UGB though the supply voltage is only 1.2 V.

Performance comparison among SPM, DPM, RFC, I-RFC, and QS-RFC to determine the amplifier implementation suitability for wearable read-out circuits is done, as shown in Figure 27. DC gain and Power dissipation being the dire

TABLE 5 Comparison of performance metrics.

Parameters	RFC	I-RFC	QS-RFC
Supply voltage (V)	1.8	1.8	1.2
Technology (nm)	180	180	130
Bias current (μA)	300	300	245
Power consumption (μW)	540	540	1.8
Capacitive load (pF)	5	5	5.2
DC gain (dB)	62.8	76.24	78.7
Unity gain BW (MHz)	34.95	74.7	180.3
Slew rate (V/ μs)	40.6	64.05	114.6
Input referred noise (μV_{rms})	201.3	139.2	48.3
Phase margin ($^\circ$)	79.78	74.41	82.5

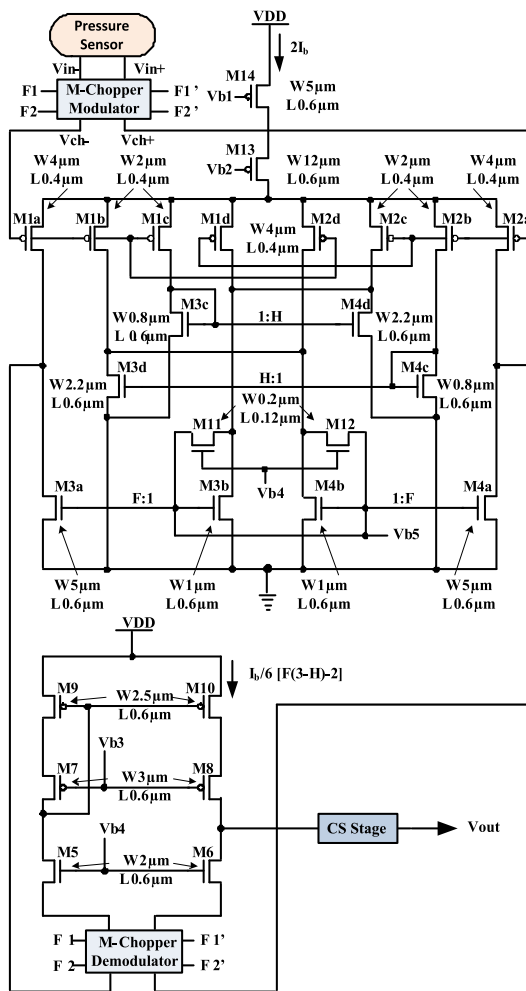


FIGURE 26 Transconductance doubling QS-RFC amplifier [44].

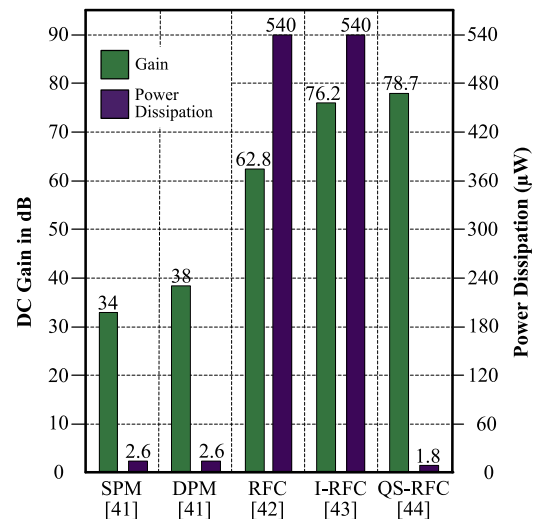


FIGURE 27 Performance comparison of RFC amplifiers.

requirement of any wearables are considered as the parameters of comparison. It is found that the QS-RFC outperformed all other designs in terms of power dissipation while yielding comparatively high DC gain.

5 | CHALLENGES OF WEARABLE TECHNOLOGIES

The challenges that prevail and dominate any wearable devices both in research and in market-ready products, especially for continuous reading/processing of vital parameters, [45] are power/battery life, data syncing/processing, signal integrity and wearability aspects. Present-day consumer tech gadgets are equipped with sophisticated fitness trackers and the associated read-out electronics for uninterrupted tracking of various parameters; however, bulky discrete interface electronics demand a high power budget. Such power demand usually requires a larger battery and hence more space within the smaller footprint of the wearable. The battery, being one of the critical and key components that keep the wearable system alive, requires a system-level stringent design to curtail power requirements [46]. In certain consumer healthcare products, though the sensor and the associated CMOS readout have a miniaturised design, there still exists a trade-off between performance and power that should be addressed efficiently by the manufacturers. To minimise the static/dynamic power dissipation, the development of power aware designs both in the MEMS sensor and as well as in the CMOS read out electronics is necessary. Minimal sensor dimensions with an appropriate choice of materials that can yield high sensitivity will enable the signal pick-up effectively without draining the battery. On the other hand, designing a high gain and low power read-out electronics and utilising the advanced CMOS technology node will be another key aspect to enhance battery performance. Such design must employ techniques such as self-biasing circuitry and capacitor-less slew rate enhancement [14] to curtail power dissipation. Additionally, the adaptation of advanced touch display technologies will enable efficient power management and hence extend battery life.

Remote health monitoring requires processing/display of reliable health data for which an enormous amount of data communication between the wearable node and the remote services is essential. Common wearables such as smart watches, smart rings, and fitness bands continuously communicate with mobile phones via communication protocols such as Bluetooth technology [47] and Bluetooth low energy technology (BLE) [48]. Further, other wearables may utilise an ARM Cortex processor and associate circuitry interfaced with the sensor to communicate with a remote PC via Zigbee [49] and Internet of Things (IoT) technology [50]. In the context of remote health monitoring applications, individual's safety, privacy, and data security are the concerns [51]. Therefore, the utilisation of state-of-the-art encryption standards will add additional faith to the system. Also, special algorithms are required to minimise the effect of noise [52, 53] to safeguard data corruption that might lead to false diagnosis.

Another key challenge is the processing and classification of health data, which are acquired from wearable nodes [54]. Different manufacturers employ distinct algorithms to realise the data and estimate the health parameters of the user. Bringing user data from a diversified range of wearables (with their consent) under one roof for better analysis demands the development of standardised algorithms that can yield high-end products. Though most microsensors discussed in the earlier sections were designed cautiously to achieve significant sensitivity while maintaining the ergonomics and comfort of the user, the noise immunity of the interface electronics still needs attention to meet the signal integrity requirement of any continuous health monitoring applications. Furthermore, designing a waterproof wearable to comply with all wearability aspects is essential. The device placed over the individual's body may get exposed to sweat, rain, or other water sources that can eventually damage the wearable device [55]. In conclusion, addressing few specific design improvements to develop a good wearable micro-sensor system is essential. Performances in terms of microsensor's sensitivity and linearity, CMOS electronics' gain and power budget, health data security and as well as robustness to harsh environment still remain inadequate for health monitoring wearables.

6 | CONCLUSION

Wearable devices that are in research both at the academic and industry level are comprehensively reviewed with suggestions on their specific applications. Few wearable devices manufactured as products by the leading semiconductor industries were also presented with the suitability of use cases and the possibility of range of applications. Further, a couple of wearable micro-systems that are available for direct usage by health care and individuals are also reviewed in depth with its performance comparisons. Based on the application targeted, novel material combinations may be explored for designing MEMS sensors to achieve maximum performance. The analog front-end circuitry that plays a critical role in determining the overall performance of the micro-sensor system has also been extensively reviewed. The proper choice and design of the read-out electronics that enable us to overcome the challenges faced in the design of modern wearable systems were also explored. Finally, the article is concluded with a discussion on the state-of-the-art CMOS + MEMS single-chip design proposed in ref. [44], which demonstrated the possibility of an on-chip monolithic integration of a MEMS capacitive pressure sensor with the advanced QS-RFC amplifier design. Such designs not only meets the power budget and noise level requirement but also might provide reliability for a continuous measurement of vital parameters even during movement, especially in critical health monitoring applications.

AUTHOR CONTRIBUTIONS

S. S. Kiran Kolluri: Data curation; investigation; validation; visualisation. **S. Ananiah Durai:** Conceptualisation; methodology; writing – reviewing and editing.

CONFLICT OF INTEREST STATEMENT

The authors have no Conflict of Interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analysed in this study.

ORCID

S. Ananiah Durai  <https://orcid.org/0000-0003-1062-7782>

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